Curated to Win: Certifying Reward-Model and Judge Selection Against Evaluation-Set Curation

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Abstract

Automated research and self-improvement pipelines choose a reward model or LLM judge by its accuracy on a validation set, and the pipeline or its designers often decide which items that set contains. We ask how much of the validation set could be removed before a different model wins, and call the answer a *curation certificate*. For benchmarks scored by weighted accuracy, the certificate has a closed form. In benchmark weight, it is at least the score gap between the top two models. In items, it is computed exactly by sorting per-item score differences. On RewardBench (118 models, 2985 items) and RewardBench 2 (188 models, 1753 items), linear programs over item weights, checked on real item subsets, show that the weight bound is nearly tight for the top models. RewardBench's winner stays first when any 11 items are removed, but removing 12 particular items puts the runner-up first. RewardBench 2's winner stays first when any 42 items are removed. If up to 1% of RewardBench's weight may have been curated away, any of 3 models could be the winner. The certificate also works on small validation sets: with 200 random items, selections whose top-two gap exceeds 2 points match the full-benchmark winner 87% of the time, against 43% for the rest. We give a selection protocol that reports the certificate, or the set of possible winners when the certificate is smaller than the curation budget.

1 Introduction

Systems that automate research or improve themselves need an evaluator: a reward model to optimize against, or an LLM judge that ranks candidate outputs (Lu et al., 2024; Zheng et al., 2023). Evaluators are chosen by meta-evaluation: each candidate is scored by how often it agrees with human preference labels on a benchmark such as RewardBench (Lambert et al., 2024; Malik et al., 2025), and the top scorer is selected. In an automated pipeline, the pipeline itself often picks the items used for this choice, as a filtered subset, a domain slice or a small validation set. If the ranking of evaluators depends on which items are kept, then whoever picks the items also picks the evaluator. This is reward hacking one level above the reward (Gao et al., 2022). A selection should therefore state how much curation it can survive.

Leaderboard rankings are known to shift with the choice of benchmark, the answer format and private testing (Dehghani et al., 2021; Alzahrani et al., 2024; Singh et al., 2025; Zhang and Hardt, 2024), and with targeted edits to individual items (Dubrovsky et al., 2025). Gordienko et al. (2026) count how many *datasets* a developer must train on to top a multi-task leaderboard, and show that this problem is NP-hard under mean win rate. Oyarhoseini et al. (2026) use influence approximations to show that editing under 1% of pairwise matches can change the leader of a Bradley–Terry arena. Both papers study how a ranking can be attacked. We take the other side: we give whoever selects an evaluator a certificate they can compute and report, for the weighted-accuracy scores used to meta-evaluate reward models and judges. Our contributions are:

- Two curation certificates for the winner of a weighted-accuracy benchmark: a lower bound in benchmark weight and an exact count in items (Props. 1–2).
- Evidence from linear programs on RewardBench and RewardBench 2 that the weight bound is nearly tight for the top models of both leaderboards.
- Two uses: the set of possible winners at a given curation budget, and certificates on small validation sets, which separate selections that hold up from those that do not.

2 Setup

A benchmark has items $i = 1, \dots, N$ with official weights $w_i^0 \geq 0$, $\sum_i w_i^0 = 1$. Model m has per-item score $r_{mi} \in [0, 1]$ and official score $S_m = \sum_i w_i^0 r_{mi}$. A *curation* keeps a subset K of the items, with kept share $f = \sum_{i \in K} w_i^0$, and rescores every model on K alone: $S_m^K = \sum_{i \in K} w_i^0 r_{mi} / f$. We call $1 - f$ the *removed share*. K *crowns* m if m scores strictly highest on it, $S_m^K > S_k^K$ for every $k \neq m$. The winner is *certified* against a class of curations if it stays first under every curation in the class.

Data. RewardBench v1 publishes per-item results for 118 models on 2985 preference pairs in 23 subsets (Lambert et al., 2024). We rebuild each official score from them: a section score is the count-weighted mean of its subsets' accuracies, with `math-prm` upweighted as in the released code, and the overall score is the mean of the four section scores. For four models, from the winner down to a 0.5B chat model, our subset accuracies match RewardBench's published values exactly. The result files list items by position, and a check of the item text found no misaligned items. RewardBench 2 (Malik et al., 2025) publishes results by item id. We use the 188 models with results on all 1753 items of its five accuracy subsets, score each model by the mean of its subset accuracies, and exclude the Ties subset, which uses a different metric.

3 Curation Certificates

Let a be the winner. For each challenger c, let $g_c = S_a - S_c > 0$ be its gap to the winner and $d_i^c = w_i^0 (r_{ai} - r_{ci})$ the contribution of item i to that gap.

Proposition 1. *Any curation K on which c ties or beats a ($S_c^K \geq S_a^K$) removes a share of at least g_c. Hence the winner is certified against every curation that removes a share below $g^\star = \min_c g_c$, its gap to the runner-up.*

Proof. $f\,(S_a^K - S_c^K) = \sum_{i \in K} d_i^c = g_c - \sum_{i \notin K} d_i^c \geq g_c - (1 - f)$, since $d_i^c \leq w_i^0$; this is positive whenever $1 - f < g_c$. □

Proposition 2. *Sort the contributions so that $d_{(1)}^c \geq d_{(2)}^c \geq \cdots$, and let $k_c = \min\{k : \sum_{j \leq k} d_{(j)}^c > g_c\}$. Removing a set R of items makes c overtake a if and only if $\sum_{i \in R} d_i^c > g_c$. So no removal of fewer than k_c items lets c overtake a, and removing the k_c items with the largest contributions does. With $k^\star = \min_c k_c$, no challenger overtakes the winner when any $k^\star - 1$ items are removed, and some $k^\star$ items let one do so, so this count is exact.*

Proof. From the identity above, $S_c^K > S_a^K$ iff $\sum_{i \notin K} d_i^c > g_c$. Any k removed items contribute at most $\sum_{j \leq k} d_{(j)}^c$. □

Both certificates need only the table of per-item scores: $g^\star$ can be read off the leaderboard, and $k^\star$ takes one sort per challenger. Prop. 1 is only a lower bound, because removing items that favor the winner over one challenger can let a third model overtake both. To see how close the bound is, we compute for each model m the largest kept share f at which some curation crowns it. For fixed f, relaxing the subset to item weights w gives a linear program:

$$\max_{w,t} \; t \;\; \text{s.t.} \;\; \sum_i w_i (r_{mi} - r_{ki}) \geq t \;\; \forall k \neq m,$$
$$0 \leq w_i \leq w_i^0 / f, \quad \sum_i w_i = 1,$$

and m can be crowned at f if $t^* > 0$. We scan f downward from 1 and then bisect. We round each LP solution to an item subset by keeping the items at or near their upper bound, and accept the subset only if m is still strictly first after rescoring, so every reported cost comes from a real, verified subset. A *section-preserving* variant also fixes each section's total weight, so that the curated set keeps the benchmark's composition. Any verified section-preserving subset is also a valid unconstrained one, so we report the unconstrained cost as the smaller of the two. For ordinal aggregates such as mean win rate, the analogous selection problem is NP-hard (Gordienko et al., 2026); for weighted accuracy, each check is a linear program.

Possible winners. Let the curation budget δ be the largest share that may have been removed, and let $P(\delta)$ be the set of models that some curation removing at most δ can crown. By Prop. 1, $P(\delta)$ contains only models within δ of the winner, $P(\delta) \subseteq \{m : S_a - S_m \leq \delta\}$. Every model whose verified LP cost is at most δ is in $P(\delta)$. Together these bound the size of $P(\delta)$ from both sides.

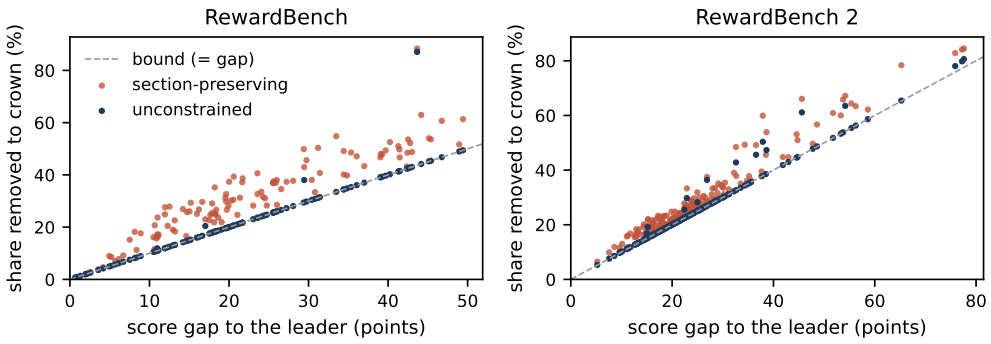


Figure 1: Share of the benchmark removed to crown each model, plotted against the model's official gap to the winner. Every point lies on or above the dashed line $y = x$, the bound of Prop. 1. Unconstrained curation reaches the bound for nearly every model; keeping each section's share fixed costs more, up to 2.31 times the gap.

4 Results

Certificates of the two winners. RewardBench's winner leads by $g^\star = 0.67$ points, so it stays first under any curation that removes less than 0.67% of benchmark weight. The cheapest crowning curation we found removes 0.68%, so the certificate is tight to within 0.01 points. In items, the winner stays first when any 11 items are removed, and removing 12 particular items makes the runner-up the new winner, which we confirmed by rescoring. (The LP minimizes weight, not item count, and its curation removed 19 items.) RewardBench 2's winner leads by 5.22 points, needs 5.30% of the weight removed to lose first place, and stays first when any 42 items are removed.

The weight bound is nearly tight. Every model on both benchmarks can be crowned by some curation. On RewardBench, the ratio of removed share to gap has a median of 1.00 and is at most 1.02 among the top 20 models (Figure 1); the largest ratio over all models is 2.00. Every top-20 model can be crowned while keeping at least 89.5% of the benchmark. On RewardBench 2, the median ratio is 1.00 and the maximum 1.35. A curator who also keeps each section's share fixed, so that the per-section breakdown shows no trace, pays a median of 1.39 times the gap on RewardBench and 1.12 on RewardBench 2.

Possible winners. Because the bound is nearly tight, the two ends of $P(\delta)$ are close (Table 1). On RewardBench, a pipeline that may have curated 1% of the benchmark can only say that the best model is one of 3, and at 5% one of 11. RewardBench 2's winner is the only possible winner up to a 5% budget.

Budget δ	0.5%	1%	2%	5%	10%
RewardBench	1	3	5–6	11	19
RewardBench 2	1	1	1	1	6

Table 1: Number of possible winners $|P(\delta)|$ at curation budget δ (share of benchmark weight). Where a range is shown, the lower end counts models with a verified LP curation within the budget, and the upper end counts models within δ of the winner (Prop. 1).

	RewardBench			RewardBench 2		
n	all	cert.	not	all	cert.	not
100	35	64 (15)	30	87	95 (76)	61
200	46	87 (8)	43	97	100 (88)	75
500	66	98 (3)	65	100	100 (97)	97

Table 2: Validation sets of n items drawn in proportion to official weight (2,000 draws per n). "all": share of draws whose selected model is the full-benchmark winner. "cert.": the same share among draws whose top-two gap on the validation set exceeds 2 points (share of such draws in brackets). "not": the same share among the other draws.

Certificates on small validation sets. In practice a pipeline scores candidates on a small validation set, where the certificate is the top-two gap on that set. Table 2 draws such sets at random, so that each set's score is an unbiased estimate of the official score. On RewardBench, a 100-item set selects the full-benchmark winner in 35% of draws. With 200 items, the 8% of selections certified at 2 points match it 87% of the time, against 43% for the rest. On RewardBench 2, whose winner leads by 5.22 points, most selections are certified and nearly all are correct. Using only the validation scores, the certificate separates selections a

pipeline can trust from those it should not.

5 A Selection Protocol

For an automated pipeline that selects an evaluator: (i) fix the validation items and record a hash of them before scoring any candidate, and log every filter applied to them; (ii) set a curation budget δ, the share of the set that could have been dropped or filtered; (iii) report the certificates $g^\star$ and $k^\star$ with the selection, and accept the winner only if $g^\star > \delta$; (iv) otherwise, report the set of possible winners, bounded as above, instead of a single winner. The certificates take one pass over the score table, so the protocol needs no extra model calls.

Limitations

We study one kind of curation, removing items, on two reward-model benchmarks. A curator who could also reweight the kept items, paraphrase them or change the prompt format would have more freedom, so the certificates hold only against removal. The item certificate is exact; the LP costs are achievable but may not be minimal. The validation-set experiment draws random sets, not adversarial ones, and uses a single threshold (2 points). We do not claim that any published result was curated. RewardBench 2's Ties subset is excluded, so our RewardBench 2 scores differ slightly from the official ones.

Reporting and Reproducibility

All results come from public per-item result files (`allenai/reward-bench-results`, `allenai/reward-bench-2-results`). All analysis ran on one laptop CPU with SciPy's HiGHS solver: each full LP sweep takes one to two hours, and the certificates take seconds. No models were run. One failed run is worth reporting. The first section-preserving search started its bisection at a very small kept share, where rounding can empty a section, and wrongly reported that the winner could not be crowned; the corrected search scans downward from $f = 1$. Code and derived data are at https://github.com/adrian-erlikhman/curated-to-win.

Use of AI Assistants

An AI assistant (Claude, Anthropic) helped write the analysis code and draft the text. The author chose the study, checked the proofs and the results, and takes responsibility for the content.

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